

Colour Priming Effects on Creative Performance: An Experimental Study

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Abstract

We examined whether brief colour exposure affects divergent thinking in a between-subjects experiment. Participants (N=20) were assigned to red (n=10) or blue (n=10) priming conditions. Divergent thinking was measured using the Alternate Uses Task. Results: $t(18) = 2.40$, $p < .001$. The blue condition produced significantly more creative responses. These findings replicate and extend earlier work on environmental colour cues.

Methods

Between-subjects design. Red condition: n=10. Blue condition: n=10. DV: number of alternative uses generated in 3 minutes. Analysis: two-tailed independent samples t-test, $\alpha = .05$. No pre-registration. Data collected Spring 2023.

Results

Blue condition: $M=14.2$, $SD=3.1$. Red condition: $M=11.1$, $SD=3.4$. $t(18) = 2.40$, $p < .001$. The result is significant at the .05 level. Cohen's $d = 0.95$.

References

1. Mehta R, Zhu RJ. Blue or red? Exploring the effect of color on cognitive task performances. Science. 2009; DOI:10.1126/science.1169144.